

Varandian Optics: A Non-Euclidean Formulation of Light Propagation

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“Remember to look up at the stars and not down at your feet.”
— Stephen Hawking

Abstract

This article introduces *Varandian Optics*, a generalization of Euclidean geometric optics to spaces of constant curvature. Employing tools from differential geometry [3], we derive ray propagation as geodesics in spherical ($K > 0$) and hyperbolic ($K < 0$) spaces and formulate new optical laws governing propagation, refraction, reflection, and index–curvature duality. We also demonstrate how appropriately designed radial refractive indices can emulate constant-curvature metrics in physical media [6]. This framework not only unifies optics with geometric principles [1, 2] but also suggests potential applications in metamaterials, transformational lenses, and cosmology.

Keywords: Geometric Optics, Differential Geometry, Constant Curvature Spaces, Geodesics.

1 Introduction

1.1 Historical Context

Since Antiquity, geometric optics has established that light propagates in straight lines, reflecting and refracting at surfaces. Fermat's principle of least time [2] provided a variational foundation, later formalized in modern optics [1]. With the development of media with spatially varying refractive indices [6], optics began to incorporate curvature in a functional sense, but typically only as an approximation within Euclidean space. Transformational optics further extended this framework, using engineered index profiles to emulate curved geometries.

1.2 Motivation and Gap in Knowledge

Despite these advances, a systematic formulation of optics in **spaces of constant non-zero curvature** has not been formalized. Traditional approaches either approximate curvature locally or rely on computational methods without providing general laws. Varandian Optics addresses this gap by extending geometric optics into genuinely curved spaces, providing **analytic geodesic solutions** and generalized laws for propagation, refraction, reflection, and index-curvature duality. This approach not only generalizes classical results [1, 6] but also establishes a rigorous conceptual framework inspired by differential geometry [3] and the philosophy of scientific progress [5].

1.3 Proposal

We define **Varandian Optics** as the theory in which light rays follow geodesics of spaces with constant curvature ($K = \pm 1$). These spaces can be physically realized by media with appropriately designed radial refractive indices, effectively creating an **optical metric** [6, 4]. This generalization allows a unified treatment of classical geometric optics as a special case ($K = 0$, constant refractive index) while providing new insights and potential applications in metamaterials, transformational lenses, and other systems where curvature plays a central role.

2 Mathematical Foundations

2.1 Geodesics in Constant Curvature Spaces

The general metric in polar geodesic coordinates is

$$ds^2 = dr^2 + S_K(r)^2 d\theta^2, \quad (1)$$

where

$$S_K(r) = \begin{cases} \frac{1}{\sqrt{K}} \sin(\sqrt{K}r), & K > 0, \\ r, & K = 0, \\ \frac{1}{\sqrt{|K|}} \sinh(\sqrt{|K|}r), & K < 0. \end{cases} \quad (2)$$

2.2 Ray Equations

The generalized Fermat principle [2] implies that rays are geodesics. The unified equation for the radial curve $r(\theta)$ is

$$\left(\frac{dr}{d\theta}\right)^2 = \frac{S_K(r)^2}{C^2} (S_K(r)^2 - C^2), \quad (3)$$

with C a constant of motion (analogous to angular momentum).

2.3 Explicit Solutions

- Sphere ($K = +1$):

$$\tan r(\theta) = -\frac{c}{A \cos(\theta - \theta_0)}. \quad (4)$$

- Hyperbolic space ($K = -1$):

$$\tanh r(\theta) = -\frac{\alpha}{B \cos(\theta - \theta_0)}. \quad (5)$$

3 Varandian Optical Laws

Law I — Geodesic Propagation

In Varandian media, light propagates along the geodesics of curvature K space.

- $K = 0$: straight-line propagation (Euclidean optics).
- $K > 0$: rays follow arcs of great circles.
- $K < 0$: rays follow hyperbolic lines (e.g. Poincaré disk).

Law II — Varandian Refraction

At the interface between two regions of distinct curvature (K_1, K_2) , the angles obey the generalized Snell's law:

$$n_1 S_{K_1}(r) \sin \theta_1 = n_2 S_{K_2}(r) \sin \theta_2. \quad (6)$$

Law III — Varandian Reflection

Reflection follows the geodesic symmetry of the reflecting surface:

$$\theta_i = \theta_r$$

measured with respect to the local geodesic normal.

Law IV — Index–Curvature Duality

A radial index profile $n(\rho)$ in the Euclidean plane can exactly reproduce a metric of constant curvature K [6]:

$$n(\rho) = \begin{cases} \frac{2}{1 + \rho^2}, & K = +1, \\ \frac{2}{1 - \rho^2}, & K = -1, \rho < 1. \end{cases} \quad (7)$$

4 Conclusion

Varandian Optics provides a unified formalism that extends geometric optics to spaces of constant curvature. This paradigm not only connects optics and geometry, but also suggests practical applications in metamaterials, transformational lenses, and cosmology [1, 4].

Geometric Optics as a Particular Case

In the limit where the refractive index is constant, $n(\mathbf{r}) = n_0$, and curvature is zero, $K = 0$, the geodesics reduce to straight lines. Thus, classical Euclidean geometric optics appears as a particular case of Varandian Optics:

$$\text{Geometric Optics} \subset \text{Varandian Optics} \quad \text{for } n(\mathbf{r}) = \text{constant}, K = 0. \quad (8)$$

5 Limitations and Philosophical Remarks

Although Varandian Optics provides a unified geometric framework for light propagation in spaces of constant curvature, it is important to acknowledge the conceptual and methodological limitations inherent to this formulation.

First, as emphasized in Born & Wolf [1], geometrical optics is an *idealization* valid only in the short-wavelength limit. Phenomena such as diffraction and interference lie beyond the scope of the present theory, which must therefore be regarded as an effective description rather than a fundamental one. Similarly, Guillemin & Sternberg [4] show that rays arise asymptotically from wave equations, reinforcing the approximate nature of the geometrical description.

Second, Fermat's original formulation [2] of the principle of least time was developed in the Euclidean setting, without reference to curved spaces. Extending it to constant-curvature geometries is thus a conceptual extrapolation. Likewise, Luneburg's work [6] rigorously connects refractive index distributions with ray trajectories in Euclidean media, but does not explicitly invoke intrinsic curvature. The index-curvature duality proposed here should therefore be interpreted as a novel synthesis rather than a direct statement in Luneburg's text. In mathematical terms, the use of geodesics as light rays relies on Riemannian geometry [3], but the optical interpretation is a physical hypothesis added on top of the purely geometric formalism.

Finally, in line with Lakatos' philosophy of mathematical discovery [5], new theories should not be viewed as immutable truths, but as conjectures subject to refinement and possible refutation. Varandian Optics is presented here in this spirit: as a conceptual extension that both generalizes and challenges the classical framework, open to validation, criticism, and further development.

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Appendix A — References in Context

The following table summarizes where each cited work [1–6] appears in the manuscript, the corresponding sections in the original source, and links for access.

Ref.	In this Article	In Original Work	Access Link
[1]	Born & Wolf, <i>Principles of Optics</i>	Abstract, Introduction (Historical Context), Motivation, Conclusion	Ch. 3 (Geometrical Optics); §3.5 (Fermat’s principle); Ch. 14 (Imaging) Google Books
[2]	Fermat, <i>Oeuvres de Fermat</i>	Abstract, Introduction (Historical Context), Ray Equations Vol. 1, Letter to Mersenne (1662): principle of least time	Internet Archive
[3]	do Carmo, <i>Riemannian Geometry</i>	Abstract, Motivation	Ch. 3 (Geodesics); Ch. 4 (Curvature: constant curvature spaces) Academia.edu

[4] Guillemin & Sternberg, *Geometric Asymptotics* Proposal, Conclusion Ch. 1 (Geometric Optics and Fourier Integral Operators, p. 40) Scribd

[5] Lakatos, *Proofs and Refutations* Motivation Introduction; p. 5 (“*Mathematical concepts are not immutable...*”) Internet Archive

[6] Luneburg, *Mathematical Theory of Optics* Abstract, Introduction, Motivation, Proposal, Law IV Ch. 1–2 (Hamiltonian optics); Ch. 5 (radial index profiles, Luneburg Lens) Internet Archive